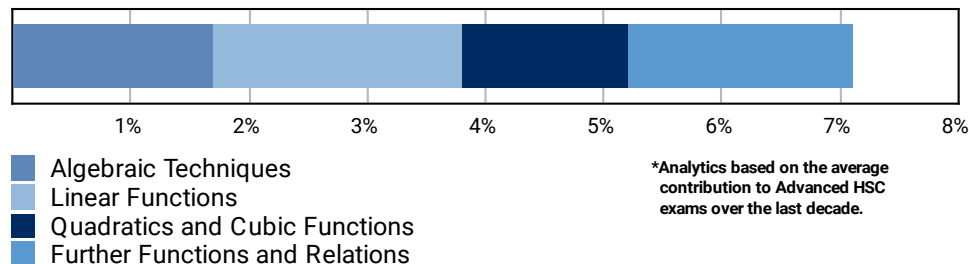


ADV: Functions (Adv), F1 Working with Functions (Adv) Linear Functions (Y11)

Teacher: Denis Vlismas

Exam Equivalent Time: 90 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

F1 Working With Functions



HISTORICAL CONTRIBUTION

- *F1 Working with Functions* is a Year 11 topic whose content represents the lowest of low hanging fruit in the new *Advanced* course.
- *F1 Working with Functions* includes new and a significant re-categorisation of old syllabus content. Any insights from past contributions in this topic area are unavoidably limited. However, with the information available, our analysis has it accounting for ~7.1%.
- We have split the topic into 5 categories for analysis purposes: 1-Algebraic Techniques, 2-Linear Functions, 3-Quadratics and Cubic Functions, 4-Composite Functions and 5-Further Functions and Relations.
- This analysis looks at *Linear Functions* (2.1%).

HSC ANALYSIS - What to expect and common pitfalls

- *Linear Functions* will primarily look at analysing and modelling linear relationships, with a significant shift away from the old course *Plane Geometry* content.
- This area represents common Adv/Std2 content and the 2020 HSC exams included a 4-mark common question (2020 Adv 11).
- Modelling and analysis of linear equations looks at numerous Std2 past HSC questions which proved challenging, including breakeven analysis.
- Although co-ordinate geometry content is diminished, some important areas remain examinable in this area and should be reviewed.
- NESA's HSC sample exam included a challenging simultaneous equation question. Review *F1 EQ-Bank 12* to cover this area.

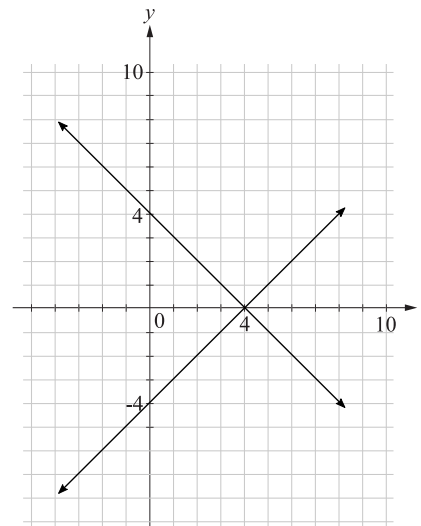
Questions

1. Algebra, STD2 A4 SM-Bank 6 MC

A computer application was used to draw the graphs of the equations

$$x - y = 4 \text{ and } x + y = 4$$

Part of the screen is shown.



What is the solution when the equations are solved simultaneously?

- A. $x = 4, y = 4$
- B. $x = 4, y = 0$
- C. $x = 0, y = 4$
- D. $x = 0, y = -4$

2. Functions, 2ADV F1 2015 HSC 2 MC

What is the slope of the line with equation $2x - 4y + 3 = 0$?

- (A) -2
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) 2

3. Functions, 2ADV F1 2017 HSC 1 MC

What is the gradient of the line $2x + 3y + 4 = 0$?

- (A) $-\frac{2}{3}$
 (B) $\frac{2}{3}$
 (C) $-\frac{3}{2}$
 (D) $\frac{3}{2}$

4. Functions, 2ADV F1 2018 HSC 3 MC

What is the x -intercept of the line $x + 3y + 6 = 0$?

- (A) $(-6, 0)$
 (B) $(6, 0)$
 (C) $(0, -2)$
 (D) $(0, 2)$

5. Algebra, STD2 A2 2019 HSC 14 MC

Last Saturday, Luke had 165 followers on social media. Rhys had 537 followers. On average, Luke gains another 3 followers per day and Rhys loses 2 followers per day.

If x represents the number of days since last Saturday and y represents the number of followers, which pair of equations model this situation?

- A. Luke: $y = 165x + 3$
 Rhys: $y = 537x - 2$
 B. Luke: $y = 165 + 3x$
 Rhys: $y = 537 - 2x$
 C. Luke: $y = 3x + 165$
 Rhys: $y = 2x - 537$
 D. Luke: $y = 3 + 165x$
 Rhys: $y = 2 - 537x$

6. Functions, 2ADV F1 2014 HSC 5 MC

Which equation represents the line perpendicular to $2x - 3y = 8$, passing through the point $(2, 0)$?

- (A) $3x + 2y = 4$
 (B) $3x + 2y = 6$
 (C) $3x - 2y = -4$
 (D) $3x - 2y = 6$

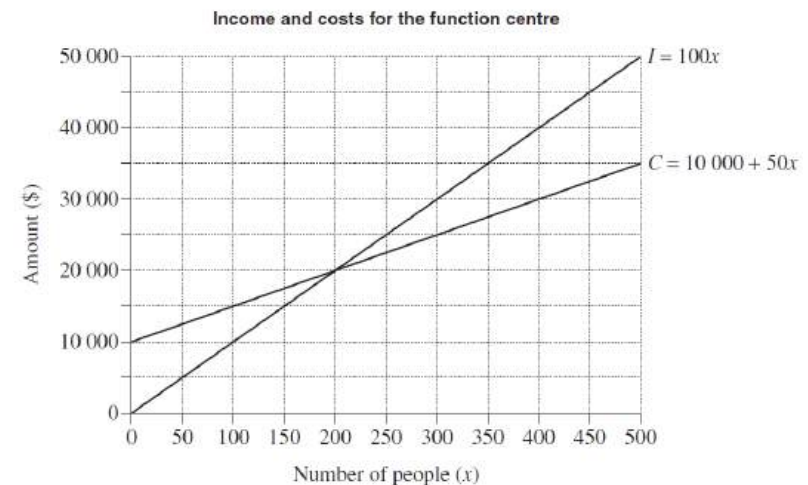
7. Algebra, STD2 A4 2011 HSC 20 MC

A function centre hosts events for up to 500 people. The cost C , in dollars, for the centre to host an event, where x people attend, is given by:

$$C = 10\,000 + 50x$$

The centre charges \$100 per person. Its income I , in dollars, is given by:

$$I = 100x$$

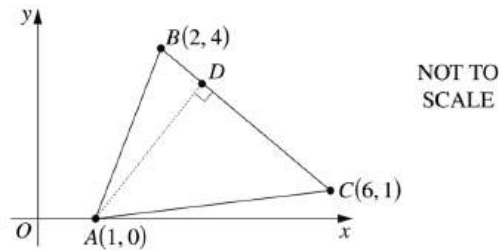


How much greater is the income of the function centre when 500 people attend an event, than its income at the breakeven point?

- (A) \$15 000
 (B) \$20 000
 (C) \$30 000
 (D) \$40 000

8. Functions, 2ADV F1 2016 HSC 12a

The diagram shows points $A(1, 0)$, $B(2, 4)$ and $C(6, 1)$. The point D lies on BC such that $AD \perp BC$.



i. Show that the equation of BC is $3x + 4y - 22 = 0$. (2 marks)

NB. Parts ii-iii are not in the new syllabus.

9. Functions, 2ADV F1 2015 HSC 11a

Simplify $4x - (8 - 6x)$. (1 mark)

10. Functions, 2ADV F1 SM-Bank 24

Ita publishes and sells calendars for \$25 each. The cost of producing the calendars is \$8 each plus a set up cost of \$5950.

How many calendars does Ita need to sell to breakeven? (2 marks)

11. Functions, 2ADV F1 2007 HSC 1f

Find the equation of the line that passes through the point $(-1, 3)$ and is perpendicular to $2x + y + 4 = 0$. (2 marks)

12. Functions, 2ADV F1 2009 HSC 1a

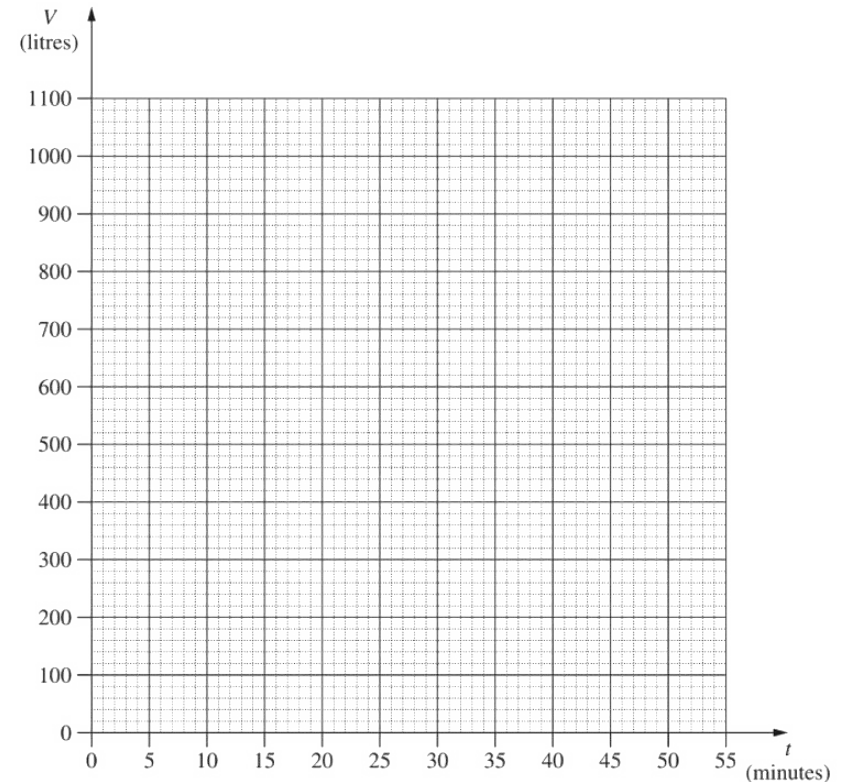
Sketch the graph of $y - 2x = 3$, showing the intercepts on both axes. (2 marks)

13. Functions, 2ADV F1 2020 HSC 11

There are two tanks on a property, Tank A and Tank B . Initially, Tank A holds 1000 litres of water and Tank B is empty.

- a. Tank A begins to lose water at a constant rate of 20 litres per minute. The volume of water in Tank A is modelled by $V = 1000 - 20t$ where V is the volume in litres and t is the time in minutes from when the tank begins to lose water. (1 mark)

On the grid below, draw the graph of this model and label it as Tank A .



- b. Tank B remains empty until $t = 15$ when water is added to it at a constant rate of 30 litres per minute.

By drawing a line on the grid (above), or otherwise, find the value of t when the two tanks contain the same volume of water. (2 marks)

- c. Using the graphs drawn, or otherwise, find the value of t (where $t > 0$) when the total volume of water in the two tanks is 1000 litres. (1 mark)

14. Functions, 2ADV F1 SM-Bank 25

Damon owns a swim school and purchased a new pool pump for \$3250.

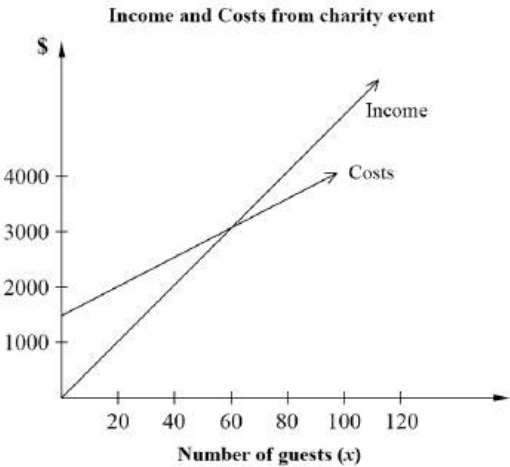
He writes down the value of the pool pump by 8% of the original price each year.

- i. Construct a function to represent the value of the pool pump after t years. (1 mark)
- ii. Draw the graph of the function and state its domain and range. (2 marks)

15. Functions, STD2 A4 SM-Bank 27

Fiona and John are planning to hold a fund-raising event for cancer research. They can hire a function room for \$650 and a band for \$850. Drinks will cost them \$25 per person.

- i. Write a formula for the cost (\$C) of holding the charity event for x people. (1 mark)
- ii. The graph below shows the planned income and costs if they charge \$50 per ticket. Estimate the number of guests they need to break even. (1 mark)



- iii. How much profit will Fiona and John make if 80 people attend their event? (1 mark)

16. Functions, 2ADV F1 EQ-Bank 12

Two archers play a game where each can aim for a large target or a small target.

If an arrow hits the large target it scores L points, and if it hits the small target, it scores S points.

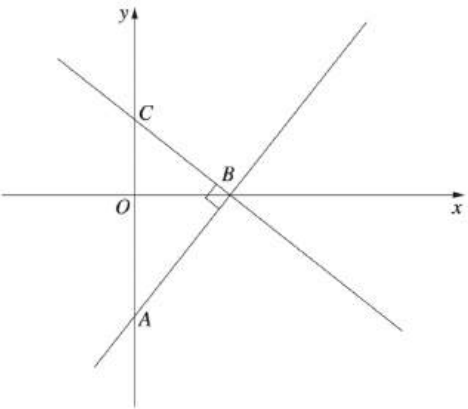
The results of a game are shown in the table below.

Archer	Number of large target hits	Number of small target hits	Total Score
1	5	8	71
2	12	5	71

By forming a pair of simultaneous equations, or otherwise, find the value of L and S . (3 marks)

17. Functions, 2ADV F1 2009 HSC 5a

In the diagram, the points A and C lie on the y -axis and the point B lies on the x -axis. The line AB has equation $y = \sqrt{3}x - 3$. The line BC is perpendicular to AB .



- i. Find the equation of the line BC . (2 marks)
- ii. Find the area of the triangle ABC . (2 marks)

18. Algebra, STD2 A2 2007 HSC 27b

A clubhouse uses four long-life light globes for five hours every night of the year. The purchase price of each light globe is \$6.00 and they each cost $\$d$ per hour to run.

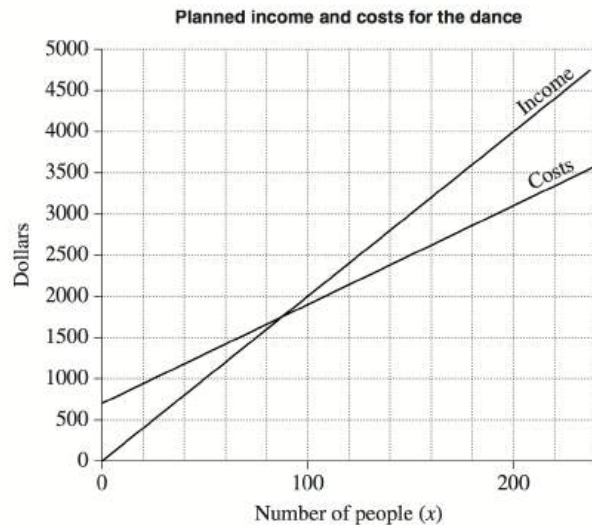
- Write an equation for the total cost ($\$c$) of purchasing and running these four light globes for one year in terms of d . (2 marks)
- Find the value of d (correct to three decimal places) if the total cost of running these four light globes for one year is \$250. (1 mark)
- If the use of the light globes increases to ten hours per night every night of the year, does the total cost double? Justify your answer with appropriate calculations. (1 mark)
- The manufacturer's specifications state that the expected life of the light globes is normally distributed with a standard deviation of 170 hours.
What is the mean life, in hours, of these light globes if 97.5% will last up to 5000 hours? (1 mark)

19. Algebra, STD2 A4 2005 HSC 28b

Sue and Mikey are planning a fund-raising dance. They can hire a hall for \$400 and a band for \$300. Refreshments will cost them \$12 per person.

- Write a formula for the cost ($\$C$) of running the dance for x people. (1 mark)

The graph shows planned income and costs when the ticket price is \$20

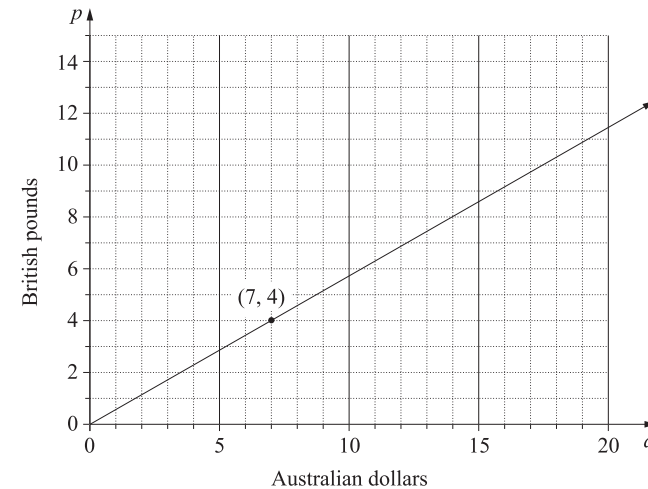


- Estimate the minimum number of people needed at the dance to cover the costs. (1 mark)
- How much profit will be made if 150 people attend the dance? (1 mark)
- Sue and Mikey plan to sell 200 tickets. They want to make a profit of \$1500.
- What should be the price of a ticket, assuming all 200 tickets will be sold? (3 marks)

20. Algebra, STD2 A2 2019 HSC 34

Part a: RAP Data - Bottom 23%: School result (19%) was -23% below state average (42%)

The relationship between British pounds (p) and Australian dollars (d) on a particular day is shown in the graph.



- Write the direct variation equation relating British pounds to Australian dollars in the form $p = md$. Leave m as a fraction. (1 mark)
- The relationship between Japanese yen (y) and Australian dollars (d) on the same day is given by the equation $y = 76d$.
Convert 93 100 Japanese yen to British pounds. (2 marks)

21. Algebra, STD2 A2 2015 HSC 27c

Ariana's parents have given her an interest-free loan of \$4800 to buy a car. She will pay them back by paying \$ x immediately and \$ y every month until she has repaid the loan in full.

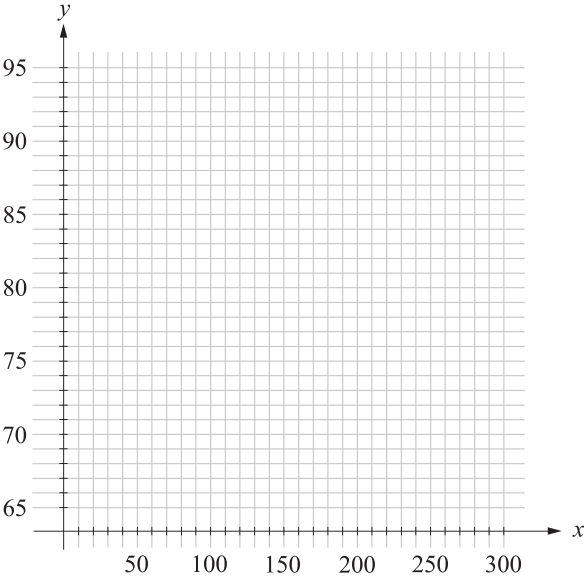
After 18 months Ariana has paid back \$1510, and after 36 months she has paid back \$2770.

This information can be represented by the following equations.

$$x + 18y = 1510$$

$$x + 36y = 2770$$

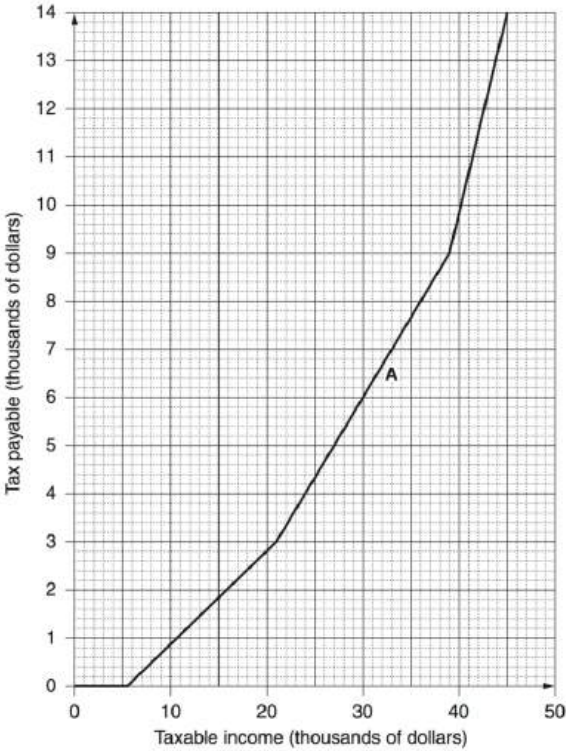
i. Graph these equations below and use to solve simultaneously for the values of x and y . (2 marks)



ii. How many months will it take Ariana to repay the loan in full? (2 marks)

22. Algebra, STD2 A2 2010 HSC 27c

The graph shows tax payable against taxable income, in thousands of dollars.



i. Use the graph to find the tax payable on a taxable income of \$21 000. (1 mark)

ii. Use suitable points from the graph to show that the gradient of the section of the graph marked **A** is $\frac{1}{3}$. (1 mark)

iii. How much of each dollar earned between \$21 000 and \$39 000 is payable in tax? (1 mark)

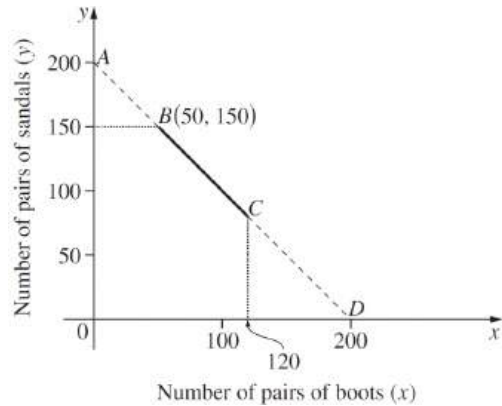
iv. Write an equation that could be used to calculate the tax payable, T , in terms of the taxable income, I , for taxable incomes between \$21 000 and \$39 000. (2 marks)

23. Algebra, STD2 A2 2009 HSC 24d

A factory makes boots and sandals. In any week

- the total number of pairs of boots and sandals that are made is 200
- the maximum number of pairs of boots made is 120
- the maximum number of pairs of sandals made is 150.

The factory manager has drawn a graph to show the numbers of pairs of boots (x) and sandals (y) that can be made.



- Find the equation of the line AD . (1 mark)
- Explain why this line is only relevant between B and C for this factory. (1 mark)
- The profit per week, $\$P$, can be found by using the equation $P = 24x + 15y$. Compare the profits at B and C . (2 marks)

Worked Solutions

1. Algebra, STD2 A4 SM-Bank 6 MC

Solution occurs at the intersection of the two lines.

$\Rightarrow B$

2. Functions, 2ADV F1 2015 HSC 2 MC

$$2x - 4y + 3 = 0$$

$$4y = 2x + 3$$

$$y = \frac{1}{2}x + \frac{3}{4}$$

$$\therefore \text{Slope} = \frac{1}{2}$$

$\Rightarrow C$

3. Functions, 2ADV F1 2017 HSC 1 MC

$$2x + 3y + 4 = 0$$

$$3y = -2x - 4$$

$$y = -\frac{2}{3}x - \frac{4}{3}$$

$$\therefore \text{Gradient} = -\frac{2}{3}$$

$\Rightarrow A$

4. Functions, 2ADV F1 2018 HSC 3 MC

x-intercept occurs when $y = 0$

$$x + 0 + 6 = 0$$

$$x = -6$$

$\therefore$ **x-intercept is $(-6, 0)$**

$\Rightarrow A$

5. Algebra, STD2 A2 2019 HSC 14 MC

Luke starts with 165 and adds 3 per day:

$$y = 165 + 3x$$

Rhys starts with 537 and loses 2 per day:

$$y = 537 - 2x$$

$$\Rightarrow B$$

6. Functions, 2ADV F1 2014 HSC 5 MC

$$2x - 3y = 8$$

$$3y = 2x - 8$$

$$y = \frac{2}{3}x - \frac{8}{3}$$

$$m = \frac{2}{3}$$

$$\therefore m_{\text{perp}} = -\frac{3}{2} \quad (m_1 m_2 = -1 \text{ for } \perp \text{ lines})$$

Equation of line $m = -\frac{3}{2}$ through $(2, 0)$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{3}{2}(x - 2)$$

$$y = -\frac{3}{2}x + 3$$

$$2y = -3x + 6$$

$$3x + 2y = 6$$

$$\Rightarrow B$$

7. Algebra, STD2 A4 2011 HSC 20 MC

When $x = 500$, $I = 100 \times 500 = \$50\,000$

Breakeven when $x = 200$ (from graph)

When $x = 200$, $I = 100 \times 200 = \$20\,000$

Difference = $50\,000 - 20\,000$

$$= \$30\,000$$

$$\Rightarrow C$$

8. Functions, 2ADV F1 2016 HSC 12a

a.i. $B(2, 4)$, $C(6, 1)$

$$m_{BC} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 4}{6 - 2} = -\frac{3}{4}$$

Equation of BC , $m = -\frac{3}{4}$ through $(2, 4)$,

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x - 2)$$

$$4y - 16 = -3x + 6$$

$$3x + 4y - 22 = 0 \dots \text{as required.}$$

9. Functions, 2ADV F1 2015 HSC 11a

$$4x - (8 - 6x)$$

$$= 4x - 8 + 6x$$

$$= 10x - 8$$

$$= 2(5x - 4)$$

♦ Mean mark 50%

COMMENT: Students can read the income levels directly off the graph to save time and then check with the equations given.

10. Functions, 2ADV F1 SM-Bank 24

Let x = number of calendars sold

$$\text{Cost} = 5950 + 8x$$

$$\text{Sales revenue} = 25x$$

Break-even occurs when:

$$25x = 5950 + 8x$$

$$17x = 5950$$

$$\therefore x = 350$$

11. Functions, 2ADV F1 2007 HSC 1f

$$2x + y + 4 = 0$$

$$y = -2x - 4$$

$$\Rightarrow \text{Gradient} = -2$$

$$\therefore \text{Perpendicular of gradient} = \frac{1}{2} \quad (m_1 m_2 = -1)$$

$$\text{Equation of line } m = \frac{1}{2}, \text{ through } (-1, 3)$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = \frac{1}{2}(x + 1)$$

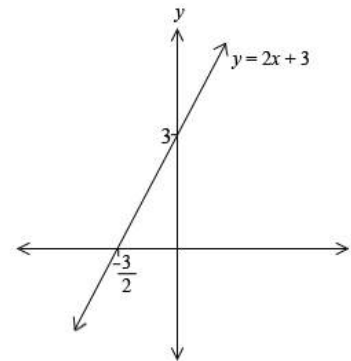
$$y = \frac{1}{2}x + \frac{7}{2}$$

$$2y = x + 7$$

$$\therefore x - 2y + 7 = 0$$

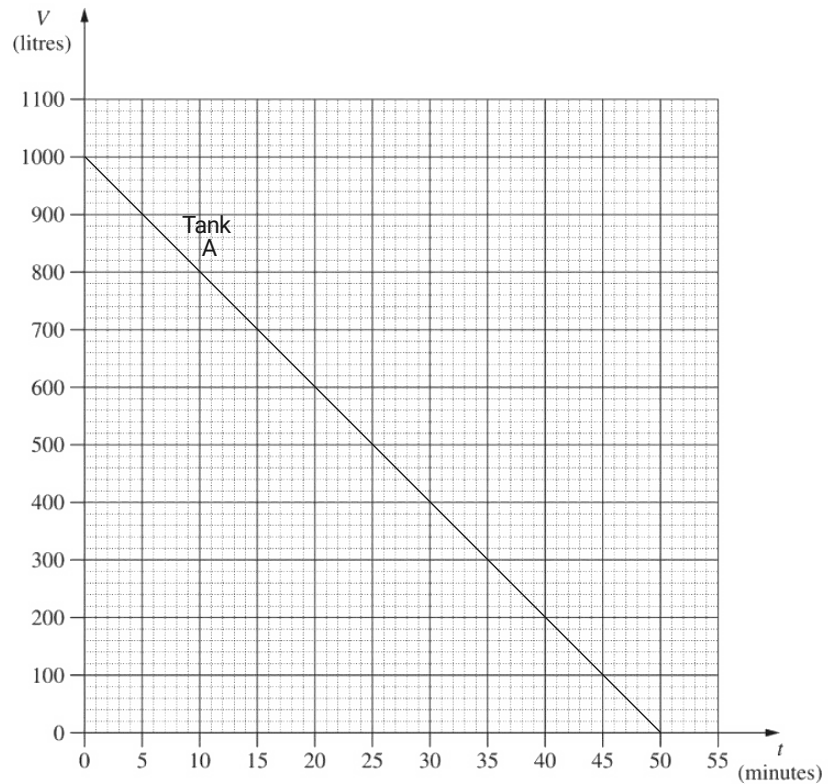
12. Functions, 2ADV F1 2009 HSC 1a

$$y - 2x = 3 \Rightarrow y = 2x + 3$$

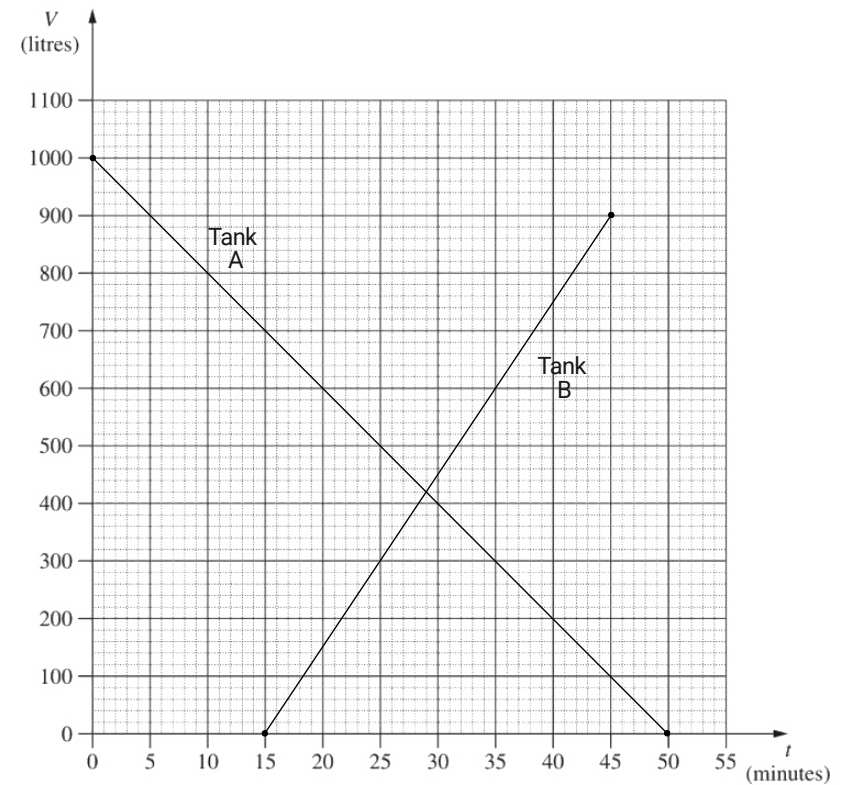


13. Functions, 2ADV F1 2020 HSC 11

a. Tank A will pass through (0, 1000) and (50, 0)



b. Tank B will pass through (15, 0) and (45, 900)



By inspection, the two graphs intersect at $t = 29$ minutes

c. Strategy 1

By inspection of the graph, consider $t = 45$

Tank A = 100 L, Tank B = 900 L

$\therefore$ Total volume = 1000 L when $t = 45$

Strategy 2

Total Volume = Tank A + Tank B

$$1000 = 1000 - 20t + (t - 15) \times 30$$

$$1000 = 1000 - 20t + 30t - 450$$

$$10t = 450$$

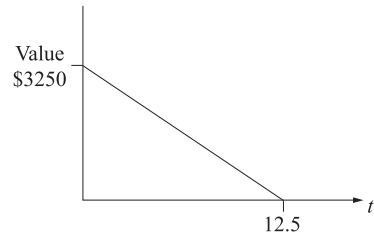
$$t = 45 \text{ minutes}$$

14. Functions, 2ADV F1 SM-Bank 25

i. Depreciation each year $= 8\% \times 3250$
 $= \$260$

$\therefore \text{Value} = 3250 - 260t$

ii.



Find t when value $= 0$

$$3250 - 260t = 0$$

$$t = \frac{3250}{260}$$

$$= 12.5 \text{ years}$$

Domain $\{t: 0 \leq t \leq 12.5\}$

Range $\{y: 0 \leq y \leq 3250\}$

15. Functions, STD2 A4 SM-Bank 27

i. Fixed Costs $= 650 + 850$
 $= \$1500$

Variable Costs $= \$25x$

$\therefore \$C = 1500 + 25x$

ii. From the graph

Costs = Income when $x = 60$

(i.e. where graphs intersect)

iii. When $x = 80$:

Income $= 80 \times 50$
 $= \$4000$

$\$C = 1500 + 25 \times 80$
 $= \$3500$

$\therefore \text{Profit} = 4000 - 3500$
 $= \$500$

16. Functions, 2ADV F1 EQ-Bank 12

$$5L + 8S = 71 \dots (1)$$

$$12L + 5S = 71 \dots (2)$$

Multiply (1) $\times$ 5

$$25L + 40S = 355 \dots (3)$$

Multiply (2) $\times$ 8

$$96L + 40S = 568 \dots (4)$$

Subtract (4) $-$ (3)

$$71L = 213$$

$$\therefore L = 3$$

Substitute $L = 3$ into (1)

$$8S = 56$$

$$\therefore S = 7$$

17. Functions, 2ADV F1 2009 HSC 5a

i. Gradient of $AB = \sqrt{3}$

$$\therefore m_{BC} = -\frac{1}{\sqrt{3}} \quad (BC \perp AB)$$

Finding B ,

$$0 = \sqrt{3}x - 3$$

$$\sqrt{3}x = 3$$

$$x = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \sqrt{3}$$

$$\therefore B(\sqrt{3}, 0)$$

Equation of BC has $m = -\frac{1}{\sqrt{3}}$ through $(\sqrt{3}, 0)$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{1}{\sqrt{3}}(x - \sqrt{3})$$

$$y = -\frac{1}{\sqrt{3}}x + 1$$

ii. AB cuts y -axis when $x = 0$, $y = -3$

$$\Rightarrow A(0, -3)$$

BC cuts y -axis when $x = 0$, $y = 1$

$$\Rightarrow C(0, 1)$$

$$\therefore AC = 4$$

$$OB = \sqrt{3}$$

$$\text{Area } \triangle ABC = \frac{1}{2} \times AC \times OB$$

$$= \frac{1}{2} \times 4 \times \sqrt{3}$$

$$= 2\sqrt{3} \text{ u}^2$$

18. Algebra, STD2 A2 2007 HSC 27b

i. Purchase price = $4 \times 6 = \$24$

$$\begin{aligned}\text{Running cost} &= \# \text{ Hours} \times \text{cost per hour} \\ &= 4 \times 5 \times 365 \times d \\ &= 7300d\end{aligned}$$

$$\therefore \$c = 24 + 7300d$$

ii. Given $\$c = \250

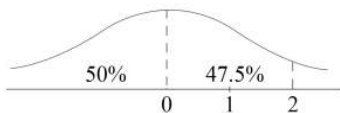
$$\begin{aligned}250 &= 24 + 7300d \\ 7300d &= 226 \\ d &= \frac{226}{7300} \\ &= 0.03095... \\ &= 0.031 \text{ \$/hr (3 d.p.)}\end{aligned}$$

iii. If d doubles to 0.062 \\$/hr

$$\begin{aligned}\$c &= 24 + 7300 \times 0.062 \\ &= \$476.60\end{aligned}$$

Since \$476.60 is less than $2 \times \$250$ (\$500), the total cost increases to less than double the original cost.

iv. $\sigma = 170$



z -score of 5000 hours = 2

$$\begin{aligned}z &= \frac{x - \mu}{\sigma} \\ 2 &= \frac{5000 - \mu}{170} \\ 340 &= 5000 - \mu \\ \mu &= 4660\end{aligned}$$

$\therefore$ The mean life of these globes is 4660 hours.

19. Algebra, STD2 A4 2005 HSC 28b

i. $\$C = 400 + 300 + (12 \times x)$
 $= 700 + 12x$

ii. Using the graph intersection

Approximately 90 people are needed to cover the costs.

iii. If 150 people attend

$$\begin{aligned}\text{Income} &= 150 \times \$20 \\ &= \$3000\end{aligned}$$

$$\begin{aligned}\text{Costs} &= 700 + (12 \times 150) \\ &= \$2500\end{aligned}$$

$$\begin{aligned}\therefore \text{Profit} &= 3000 - 2500 \\ &= \$500\end{aligned}$$

iv. Costs when $x = 200$:

$$\begin{aligned}C &= 700 + (12 \times 200) \\ &= \$3100\end{aligned}$$

Income required to make \$1500 profit

$$\begin{aligned}&= 3100 + 1500 \\ &= \$4600\end{aligned}$$

$$\begin{aligned}\therefore \text{Price per ticket} &= \frac{4600}{200} \\ &= \$23\end{aligned}$$

20. Algebra, STD2 A2 2019 HSC 34

a. $m = \frac{\text{rise}}{\text{run}} = \frac{4}{7}$
 $p = \frac{4}{7}d$

♦ Mean mark 42%.

b. Yen to Australian dollars:

$$y = 76d$$

$$93\,100 = 76d$$

$$d = \frac{93\,100}{76}$$

$$= 1225$$

Aust dollars to pounds:

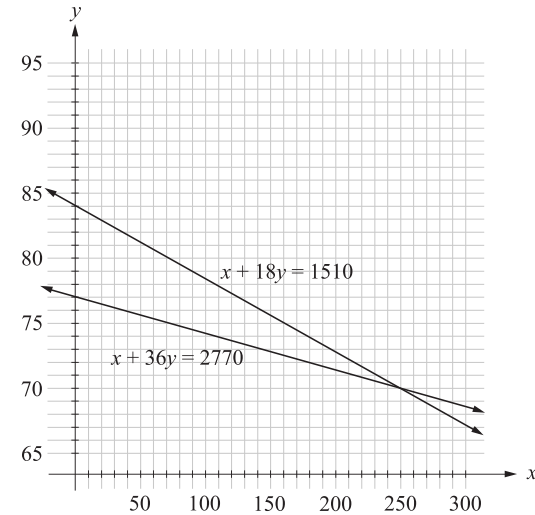
$$p = \frac{4}{7} \times 1225$$

$$= 700 \text{ pounds}$$

$$\therefore 93\,100 \text{ Yen} = 700 \text{ pounds}$$

21. Algebra, STD2 A2 2015 HSC 27c

i.



$$\therefore \text{Solution is } x = 250, y = 70$$

ii. Let A = the amount paid back after n months

$$A = 250 + 70n$$

Find n when $A = 4800$

$$250 + 70n = 4800$$

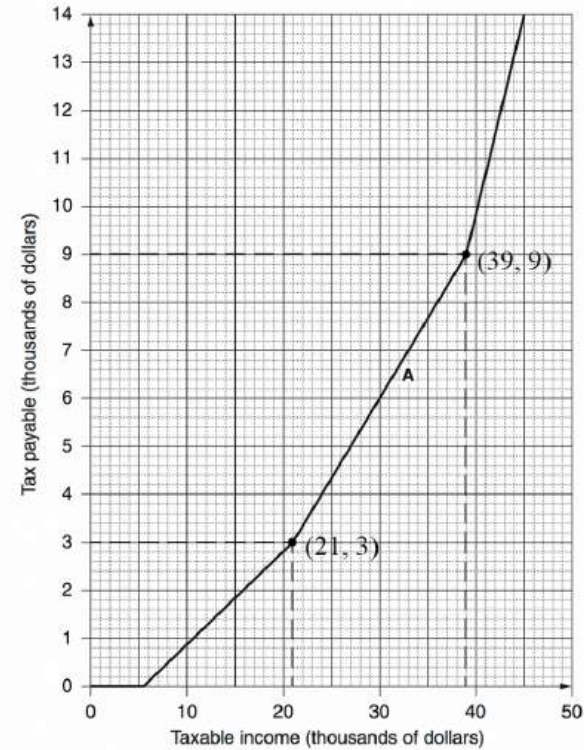
$$70n = 4550$$

$$n = 65$$

♦ Mean mark 44%.

$\therefore$ It will take Ariana 65 months to repay the loan in full.

i.



Income on \$21 000 = \$3000 (from graph)

ii. Using the points (21, 3) and (39, 9)

$$\begin{aligned}
 \text{Gradient at A} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{9000 - 3000}{39\,000 - 21\,000} \\
 &= \frac{6000}{18\,000} \\
 &= \frac{1}{3} \quad \dots \text{ as required}
 \end{aligned}$$

♦♦ Mean mark 25%

iii. The gradient represents the tax applicable to each dollar

$$\text{Tax} = \frac{1}{3} \text{ of each dollar earned}$$

♦♦♦ Mean mark 12%!
MARKER'S
COMMENT: Interpreting gradients

$$= 33\frac{1}{3} \text{ cents per dollar earned}$$

is an examiner favourite, so make sure you are confident in this area.

iv. Tax payable up to \$21 000 = \$3000

Tax payable on income between \$21 000 and \$39 000

$$= \frac{1}{3}(I - 21\,000)$$

$$\therefore \text{Tax payable on } I = 3000 + \frac{1}{3}(I - 21\,000)$$

$$= 3000 + \frac{1}{3}I - 7000$$

$$= \frac{1}{3}I - 4000$$

♦♦♦ Mean mark 15%.

STRATEGY: The earlier parts of this question direct students to the most efficient way to solve this question. Make sure earlier parts of a question are **front and centre of your mind** when devising strategy.

23. Algebra, STD2 A2 2009 HSC 24d

i. We are told the number of boots (x),

and shoes (y), made in any week = 200

$$\Rightarrow \text{Equation of } AD \text{ is } x + y = 200$$

♦♦♦ Mean mark part (i) 14%.

Using $y = mx + b$ is a less efficient but equally valid method, using $m = -1$ and $b = 200$ (y -intercept).

ii. Since the max amount of boots = 120

$$\Rightarrow x \text{ cannot } > 120$$

Since the max amount of sandals = 150

$$\Rightarrow y \text{ cannot } > 150$$

$\therefore$ The line AD is only possible between B and C .

♦ Mean mark 49%

iii. At B , $x = 50$, $y = 150$

$$\Rightarrow \$P \text{ (at } B) = 24 \times 50 + 15 \times 150$$

$$= 1200 + 2250$$

$$= \$3450$$

At C , $x = 120$, $y = 80$

$$\Rightarrow \$P \text{ (at } C) = 24 \times 120 + 15 \times 80$$

$$= 2880 + 1200$$

$$= \$4080$$

♦ Mean mark 40%.

$\therefore$ The profits at C are \$630 more than at B .